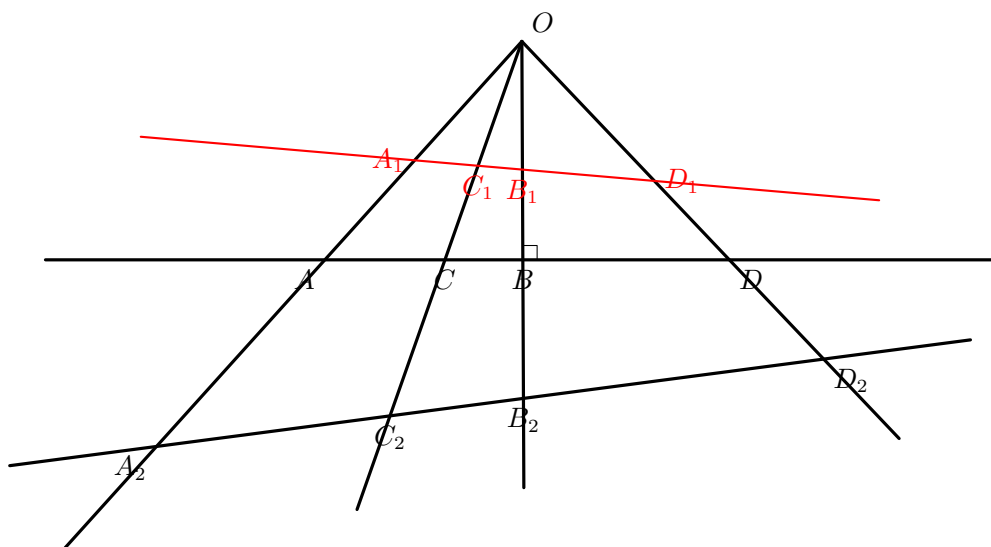


1 射影几何

文艺复兴时期，画家研究投影，研究出射影不变性

2 交比



$\left(\frac{AC}{AB}\right) / \left(\frac{BC}{BD}\right)$ 之比，称为交比。

1. 交比具有射影不变性。

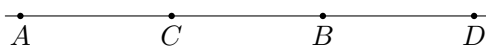
(由面积比推出的交比表达式)

$$\begin{aligned} \left(\frac{AC}{AD}\right) / \left(\frac{BC}{BD}\right) &= \left(\frac{S_{\triangle AOC}}{S_{\triangle AOD}}\right) / \left(\frac{S_{\triangle BOC}}{S_{\triangle BOD}}\right) \\ &= \left(\frac{\frac{1}{2} OA \cdot OC \cdot \sin \angle AOC}{\frac{1}{2} OA \cdot OD \cdot \sin \angle AOD}\right) / \left(\frac{\frac{1}{2} OC \cdot OB \cdot \sin \angle COB}{\frac{1}{2} OD \cdot OB \cdot \sin \angle BOD}\right) \\ &= \left(\frac{\sin \angle AOC}{\sin \angle AOD}\right) / \left(\frac{\sin \angle COB}{\sin \angle BOD}\right) = \frac{\sin \angle AOC}{\sin \angle AOD} \cdot \frac{\sin \angle BOD}{\sin \angle COB}. \end{aligned}$$

二、当交比为 -1 (调和比)

$$\overrightarrow{AC} = \lambda \overrightarrow{CB}, \quad \overrightarrow{AD} = -\lambda \overrightarrow{DB}, \quad (\lambda > 0 \text{ 且 } \lambda \neq 1),$$

$$\left|\frac{AC}{AD}\right| = \left|\frac{CB}{DB}\right|.$$



A, C, B, D 调和点列:

1. A, B 为基点, C, D 为内外分点, CD 调和分割 AB 。

2. C, D 为基点, B, A 为内外分点, AB 调和分割 CD 。

3. AB, CD 调和共轭。

$$\frac{2}{AB} = \frac{1}{AC} + \frac{1}{AD}.$$

记:

$$\frac{BC}{AC} = \frac{BD}{AD}.$$

$$\begin{aligned} \frac{AB - AC}{AC} = \frac{AD - AB}{AD} &\Rightarrow \frac{AB}{AC} - 1 = 1 - \frac{AB}{AD} \\ \Rightarrow \frac{AB}{AC} + \frac{AB}{AD} = 2 &\Rightarrow \frac{1}{AC} + \frac{1}{AD} = \frac{2}{AB}. \end{aligned}$$

调和点列 $\begin{cases} 1. \text{交比为 } -1 \text{ (调和平均)} \\ 2. \text{射影不变性} \end{cases}$

